RPI

AI-Driven Company-Wide Search Engine
for Deloitte Resources

Group Two Project Proposal

Team Members: Jesse Gabriel, Andrew Jung, Raven Levitt, Felix Tian, Sophia Turnbow, Matthew Voynovich

Course: ITWS 4100 - Information Technology Capstone

Professor: Richard Plotka

Date: February 12, 2026

Executive Summary

A second-year consultant in Chicago is preparing a client pitch on digital transformation. She needs a slide deck she saw three months ago, but cannot remember the title, the author, or which of the four internal platforms it lives on. She spends 40 minutes searching across internal resources, messages two colleagues on Teams, and eventually settles for an outdated version. Now multiply that by hundreds of thousands of professionals, every single day.

This is the core problem Deloitte's Project 2 brief identifies: employees rely on a variety of internal sites for documents, policies, slide decks and more. However, searching across these platforms is difficult and often tedious, especially when they cannot clearly describe what they need. A unified search engine powered by Natural Language Processing (NLP) that understands users' intents, delivers accurate results, and measurably improves productivity is a competitive necessity.

What We Propose

A web application with one search bar and natural language understanding. It will deliver instant, relevant results across all provided resources with clearly defined query boundaries, security guardrails, and built-in guidance that helps users find what they're looking for more effectively.

Deloitte will receive the following deliverables by the end of this project:

- **Defined use cases and requirements** grounded in interviews and data
- **Market research on NLP security and boundaries** with actionable recommendations
- **Technical architecture documentation** with diagrams addressing business logic, security, and compliance
- **A functioning web application** demonstrating intelligent search across provided Deloitte resources

The Cost of the Status Quo

External data on this problem class is worth quantifying, because it frames the return on investment of any solution:

- **62% of workers** report spending too much time searching for information during their workday (Microsoft Work Trend Index, 2023)
- **2.8 hours per week on average** spent by knowledge workers on locating or requesting needed information, contributing to a **25% loss of the average workweek** to productivity drains (APQC, 2021)
- **Only 27% of companies** have implemented effective search tools to address this issue (Slite, 2025)

At Deloitte's scale, even a modest improvement in search efficiency recovers significant billable capacity. Industry-wide, professional services billable utilization averaged **68.9%** in 2024, which is below the **70-75% optimal threshold** where profitability is maximized (SPI Research, 2025). Bridging this gap is precisely the kind of operational improvement that better tooling enables.

The Team

Our team brings direct experience from across the three disciplines this project demands: NLP-powered retrieval systems, AI safety and compliance, and user-centered design research. Rather than building from scratch, we are positioned to draw on existing architectural expertise and adapt proven approaches to Deloitte's specific needs.

Raven Levitt - *Project Manager*

Raven Levitt brings a rare combination of executive leadership and full-stack technical expertise to the Project Manager role. As a Co-founder of Meridian, Raven has successfully built and deployed software systems serving thousands of active users, proving his ability to manage the full product lifecycle from concept to launch. His operational experience is equally robust; as a VP at HackRPI and a leader within RCOS, he has directed teams ranging from agile squads of 5 to large-scale operations of over 100. This unique background allows him to seamlessly translate business requirements into technical reality, ensuring that development stays on track, on budget, and aligned with the project's strategic vision.

Felix Tian - *System Architect*

Felix Tian is a senior Information Technology & Web Science student at RPI. He is the lead developer on the FinGPT Search Agent project hosted at Columbia University. It is a retrieval-augmented search agent powered by the FinGPT model. He has published a paper on the search agent, "Customized FinGPT Search Agent Using Foundational Models", that proposes an institutional version designed for numerical accuracy, reduced hallucination, and air-gapped deployment that enables organizations to retrieve and leverage proprietary data securely. Much of the architectural groundwork for the kind of system Deloitte has described has already been developed by Felix, providing us a meaningful headstart for Deloitte's project.

Sophia Turnbow - *User Research Lead*

Sophia is a senior Information Technology & Web Science and Sustainability Studies dual major with a strong background in user experience design and user research. Her experience spans information discovery, workflow optimization, and user-centered system design across academic and applied projects. She has led qualitative research efforts including interview design, user recruitment, usability testing, and insight synthesis for data-driven digital products. As Chief Marketing Officer at Project Almanac, Sophia applies user research and systems thinking to guide product positioning, information clarity, and

adoption strategy, working closely with technical and product teams to align user needs with organizational goals.

Matthew Voynovich - *Security Researcher*

Matthew Voynovich is a researcher focused on AI agent workflow safety, privacy leakage, and policy compliance in agent orchestration frameworks such as LangChain and LangGraph. His work examines how AI agent workflows can be constrained to satisfy both regulatory requirements and internal enterprise policy controls, directly addressing risk management and AI governance challenges in deploying NLP and agent systems faced by firms such as Deloitte. He is a dual major in Computer Science and Information Technology & Web Science at Rensselaer Polytechnic Institute with a concentration in AI, machine learning, and data science. In parallel, he has contract experience as a full-stack developer, delivering production web applications using React, Vue.js, Node.js, TypeScript, and PostgreSQL, including systems that incorporate LLM-enabled functionality.

Jesse Gabriel - *Developer*

Jesse Gabriel is a senior pursuing a dual major in Computer Science and Information Technology & Web Science with a concentration in Information Security, as well as a minor in Business Management at Rensselaer Polytechnic Institute. As a former Cyber Analyst intern at Deloitte within the Government and Public Services practice, Jesse brings direct experience navigating Deloitte's internal systems, compliance expectations and consulting workflows that are relevant for this project. His security background ensures that the search engine's data handling, access controls, and query boundaries meet enterprise standards, while his direct familiarity with how Deloitte professionals interface with internal resources grounds the team's design decisions in real operational context.

Andrew Jung - *Market Researcher*

Andrew Jung is a senior Information Technology & Web Science Major with an Economics Minor at Rensselaer Polytechnic Institute. As a co-founder and Frontend Lead at TandmAI, an AI governance startup focused on preventing employees from leaking sensitive and proprietary information through generative AI tools, Andrew worked closely with policy enforcement, prompt classification, and role-based access controls designed to safeguard PII and confidential company data. Drawing on this experience, he leads market research into NLP security, privacy, and query boundaries, evaluating how modern AI-powered search systems restrict access, handle sensitive data, and enforce safe query behavior. His work translates real-world enterprise risks and best practices into actionable system requirements, helping ensure the search solution balances powerful natural-language usability with Deloitte’s security, compliance, and data governance standards.

Scope

In Scope

- Stakeholder interviews and persona development across Deloitte employees (tax, audit, and consulting) to define user needs, workflows, and search behaviors
- Market research on NLP security best practices and boundaries
- Architecture diagrams and documentation addressing business logic, security, and compliance requirements
- Formal documentation of use cases and user journeys to inform NLP modeling and feature prioritization
- User-friendly web application for natural language queries
- NLP-based intent recognition and resource matching
- Searchable database of all company-provided resources (documents, policies, slide decks, images)
- Query boundary enforcement (language support, input validation, nonsensical query handling)
- User guidance (search tips, user walk-throughs, query suggestions)

Out of Scope

- Integration with Deloitte's live production system or real-time data feeds
- Multilingual NLP support beyond English
- Mobile-native application development
- Ongoing maintenance or post-deployment support
- Access to or processing of actual confidential client data

Technical Components

The following components directly address each technical requirement outlined in the project brief.

1. A User-Centric Friendly Search Application

A single-page web application where users type natural language queries and get relevant results:

- **Search bar** accepting free-form queries (e.g., "Q4 healthcare consulting deck," "travel reimbursement policy," etc.)
- **Results display** showing ranked item with title, snippet, author, date, data type and other metadata
- **Filtering** by document type, date range, and other metadata

2. Natural Language Processing for Intent Understanding

The NLP engine interprets users' intents, and not just what they type:

- **Query understanding** via tokenization, intent classification, and entity extraction
- **Semantic search** converting queries to vector embeddings, retrieve results by combining meaning similarity with keyword matching
- **Ranking system** scores results by semantic similarity, keyword match quality, recency and metadata signals

3. Query Boundaries

The brief specifically calls for defining what queries the engine can and cannot process:

- **Language** supporting only English for the prototype
- **Input validation** enforcing length limits, sanitization of nonsensical input, blocking prompt-injection patterns and personally identifiable information (PII)
- **Content filtering** rejects queries containing profanity, offensive language, or patterns attempting to exploit NLP models
- **Rate limiting** prevents abuse
- **Graceful degradation** provides helpful error messages for empty, malformed, or zero-result queries

4. User Guidance

Built-in guidance features to reduce the learning curve and improve search effectiveness:

- **Contextual suggestions** when queries return poor results
- **Query suggestions** based on indexed content, past searches, and common search patterns
- **"Did you mean?"** corrections for typos and near-matches
- **UI Walkthrough** providing first-time users an overview on how to use the web application

5. Searchable Resource Database

Depending on Deloitte's requirements, organize all provided resources into a dual-database architecture or integrate with Deloitte's existing infrastructure:

- **Metadata store** for structured data (title, author, date, data type, permissions), faceted filtering, and audit logs
- **Vector store** for document embeddings for semantic search, optimized for low query latency
- **Indexing pipeline** to ingest documents (primarily PDFs and Word Docs, with extensibility for other formats), extract text and metadata, chunk long documents, generate embeddings and store and cross-reference via unique document IDs

6. Administrative and Governance Controls

Support oversight, system tuning, and responsible system use:

- **Configurations of supported query** types and system boundaries
- **Role-Based Access Control (RBAC)** for controlled retrieval of resources and information
- **Logging** of search activity
- **Documentation** of security considerations and system limitations

9-Week Timeline (Preliminary)

With a high estimate, there are around 9 weeks from now until handoff. A sample timeline is created for reference. Subject to change when more project details become available.

Week	Focus	Next Key Actions
Week 1	Kickoff; confirm research approach	Interview scheduling; data timeline
Week 2	Aggregate interview findings	Finish user personas and use case priorities; Start architecture design and UI design
Week 3	Research reports finished; Collect feedback of UI designs	Architecture direction review
Week 4	Technical design validated	Start coding; Finalize search result format: links only vs. links + AI summary
Week 5	Tech stack and pipeline validation	Developing the prototype
Week 6	Gain feedback on work done so far	Iterate upon the prototype based on feedback
Week 7	Beta prototype demo with UI/UX feedback	Usability testing plan review; Iterate upon the UI/UX feedback
Week 8	Usability test results	Final iteration priorities
Week 9	Final presentation and handoff	—

Technology Stack (Preliminary)

The following reflects our current direction based on limited available project details. Final selections, and the corresponding data flow diagram, will be validated with the Deloitte team during weekly check-ins as requirements become clearer, particularly once sample data are available.

Layer	Technology	Notes
Frontend	React + TypeScript	Component-based search interface and responsive design for desktop use
Backend	Python (FastAPI)	Industry-standard for AI-integrated applications. Mature ecosystem and large community
Document Processing	MarkItDown / LlamaParse / PageIndex	PDF/Word parsing tools optimized for LLMs. OCR fallback (e.g. Docling) if scanned documents are included; PageIndex to parse long documents
NLP / Embeddings	Semantic embedding model (e.g. BGE-M3)	Model selection pending evaluation against Deloitte's sample data
Search Architecture	Hybrid retrieval (semantic + keyword)	Link-based results ranked by relevance. Supports searching titles and content within resources.
Database	Relational + Vector store (e.g. PostgreSQL + Pinecone)	Specific choices to be finalized after data type and volume are known
Deployment	Docker containers	Reproducible environments and cloud-ready
Version Control	GitHub	Industry-standard platform for collaborative development, code review, documentation hosting, and more

NLP Security & Data Handling

The brief calls for market research on NLP security and boundaries for sensitive company data. Our research will cover the following points. Details may change as we obtain feedback on this research.

- Survey of enterprise search security practices across leading platforms (e.g. Glean, Elastic, Microsoft Search)
- Analysis of query-time access control patterns to gain an insight on how permission checks are enforced before results are returned

- Study of NLP-specific attack patterns: prompt injection, data exfiltration via crafted queries, embedding inversion, etc.
- Best practices for defining query processing limitations, including what the engine should and should not accept
- Compliance considerations (GDPR, SOC 2) for indexing and retrieving internal resources

This will be delivered as a standalone report with actionable recommendations.

Risk Management

The level of impact and likelihood is estimated based on available knowledge and project details. This chart may be adjusted when more details of this project become available.

Risk	Impact	Likelihood	Mitigation
Search Relevance Misses	Medium (User frustration)	Medium	Hybrid retrieval for balanced precision / recall; Query logs to identify failure patterns; "Did you mean?" suggestions
Security / Permission Violations	High (Compliance risk)	Low	Query-time RBAC enforcement; fail-closed design; Security review as dedicated deliverable; Full audit logging
Data Quality Issues	Medium (Poor search relevance)	Unknown	Content analysis to infer missing metadata; Ranking algorithm robust to gaps
Timeline Pressure (~9 weeks from now until handoff)	High (Incomplete Delivery)	Low	Parallel workstreams; Prioritized feature set (core features first, enhancement later); Weekly internal milestones
User Adoption Resistance	High (Tool goes unused)	Low	Research-driven (user-centric) design; Fast iteration of UI during design stage; Guidance features lower learning curve; Simple "Was this helpful" measuring satisfaction

The Vision

Remember the consultant in Chicago? She spent 40 minutes searching for a slide deck, messaged two colleagues, and settled for an outdated version.

That same morning looks different now: she opens this search engine, types "digital transformation client pitch deck Q3", and the first result is exactly what she needs. The

author, the date, and a preview snippet all confirm it's the right version. She clicks, downloads, and moves on, with the entire process taking under a minute. This is what we are building toward: a proof-of-concept that demonstrates what becomes possible when a search engine understands both users' intents and the complex structure of a company's internal resources. Our goal is for Deloitte to walk away from this project with a clear picture of what a production-grade version could do for hundreds of thousands of professionals, every day.

References

Microsoft. "Work Trend Index: Will AI Fix Work?" May 2023. Survey of 31,000 workers across 31 countries, combined with M365 telemetry data. https://www.microsoft.com/en-us/worklab/work-trend-index/will-ai-fix-work

APQC. "Fixing Process & Knowledge Productivity Problems: Survey Report." 2021. https://www.apqc.org/resource-library/resource-listing/fixing-process-knowledge-productivity-problems-survey-report

Service Performance Insight (SPI Research). Professional Services Maturity Benchmark Report. 2025. https://forms.workday.com/content/dam/web/en-us/documents/reports/SPI_2025_Benchmark_Report.pdf?refCamp=7014X000001yvgK

Slite. "Enterprise Search Survey Report." 2025. Survey of 100+ global knowledge workers on enterprise search adoption. https://slite.com/en/learn/enterprise-search-survey-findings